

Project

Marketing Customer Analytics

Utilized:

Python: Pandas, NumPy, Matplotlib, Seaborn, data cleaning, feature engineering, EDA.

SQL: Data validation, aggregations, GROUP BY, HAVING, CASE, subqueries, CTEs, window functions, business queries.

Power BI: Power Query, DAX, Measures, Data Modeling, KPIs, interactive charts, filters and dashboard design.

Statistics: Descriptive statistics, aggregations, correlation analysis, distribution and outlier analysis.

Created: Cleaned and transformed customer marketing data, performed exploratory analysis, answered business questions using SQL, and designed an interactive Power BI dashboard for customer, product, purchase-channel and campaign analysis.

Found Key Insights like:

- **Wines and Meat Products are the highest-spending** product categories, making them suitable for bundles, cross-selling, and premium promotions.
- **Graduation/Education group has the highest total spending**, indicating that this customer segment has strong purchasing potential.
- Monthly spending fluctuates significantly: **spending peaks in August (~145K)** and remains strong from September to November, while **July (~50K) records the lowest spending**. This suggests an opportunity for seasonal promotions during low-spending periods.
- **Store remains the leading purchase channel**, followed by Website and Catalog, supporting channel-specific marketing strategies.
- The Most Recent Campaign generated the highest response (~334), considerably higher than the other campaigns, making it useful to investigate for successful targeting strategies.
- Overall, combining education, income, age, recency, spending, and purchasing behaviour provides a strong basis for customer segmentation and targeted marketing.

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Report analysis: Marketing Customer Analytics

Project Overview

The Marketing Customer Analytics project analyzes customer-level marketing data to uncover actionable insights for improving customer targeting, campaign effectiveness, product promotion and purchase-channel strategy. The analysis combines Python-based data preparation and EDA, SQL business analysis and Power BI reporting.

Objectives

- Understand the demographic and purchasing profile of customers.
- Identify high-performing product categories and customer spending patterns.
- Evaluate web, catalog and store purchasing behavior.
- Measure marketing campaign acceptance and response.
- Identify customer groups with different value, engagement and retention characteristics.
- Provide actionable recommendations for marketing and customer strategy.

Key Metrics

- Total Customers
- Total Customer Spending
- Average Customer Spending
- Average Income
- Total Purchases
- Web, Catalog and Store Purchases
- Campaign Acceptance / Response Rate
- Average Recency

Project Workflow

1. Load and inspect the Maven Analytics dataset.
2. Clean, validate and transform the data using Python.
3. Perform descriptive and exploratory analysis to identify patterns.
4. Load the cleaned data into SQL and answer business questions.
5. Create a structured Power BI data model and DAX measures.
6. Build interactive dashboards and communicate business insights.

Customer & Product Performance Analysis

Customer Profile Analysis

- Customer demographics were analysed using age, income, education, marital status, country, and household characteristics.
- Age and income groups were compared to understand differences in customer spending.
- The number of children was analysed to examine its relationship with customer spending.
- Filters for demographic attributes allow different customer segments to be explored interactively..

Product Spending Analysis

- Spending across Wines, Fruits, Meat Products, Fish Products, Sweet Products, and Gold Products was compared.
- **Wines** contributed the highest share of product spending (50.17%), followed by Meat Products (27.56%) and Gold Products (13.46%).
- Product-level results support targeted promotions, cross-selling, and product-bundle decisions.

Purchase Channel Analysis

- Store, Website, and Catalog purchases were compared to identify preferred purchase channels.
- **Store** purchases were highest at approximately 13K, followed by Website (9K) and Catalog (6K).
- Different customer segments can be targeted with channel-specific promotions based on their purchasing preferences.

Business Interpretation

The analysis connects customer demographics, spending behavior, product preferences, and purchase channels to identify meaningful customer segments. These insights provide a foundation for targeted marketing, product promotion, customer retention, and channel-specific strategies.

Campaign & Customer Engagement Analysis

Campaign Performance

- Responses to Campaigns 1–5 and the latest campaign were compared to identify stronger and weaker campaign performance.
- The latest campaign recorded the highest response count (334), followed by Campaign 4 (167), Campaign 3 (163), Campaign 5 (163), Campaign 1 (144), and Campaign 2 (30).
- Overall campaign response was approximately 7.45%, indicating an opportunity to improve campaign targeting and engagement.

Customer Engagement

- Recency was analysed to distinguish recently active customers from customers who may be becoming inactive.
- Total Purchase and Total Spending were used to understand customer activity and value.
- Age, income, education, and number of children were analysed to understand differences in customer behaviour.

Correlation & Statistical Analysis

- Correlation analysis was used to examine relationships between income, spending, purchases, and customer engagement variables.
- Distribution analysis and outlier detection were performed during EDA to understand the characteristics of numerical variables.
- Correlation results were interpreted as relationships and not proof of causation.

High-Level Insights

- Customer value depends on spending and purchasing activity, rather than a single demographic characteristic.
- Store purchases are the highest, while Website and Catalog channels provide opportunities for further growth.
- Wines (50.17%) and Meat Products (27.56%) are the major contributors to product spending.
- Middle-aged and higher-income customers represent important segments for targeted marketing.
- Campaign performance can be improved by studying successful response patterns and applying them to future campaigns.

Business Questions & Answers

1. Which products should the business prioritize?

Wines should be the top priority, contributing 50.17% of total product spending, followed by Meat Products at 27.56%. These products should be used for bundles, cross-selling, and premium offers.

2. Which purchase channel should receive attention?

Store is the leading channel with 13K purchases, followed by Website (9K) and Catalog (6K). The business should maintain strong store promotions while improving digital channels through personalized offers.

3. Which customers are most valuable?

Customers with high total spending and strong purchase activity should be classified as high-value customers. Your analysis also shows that higher-income and middle-aged segments are important targets for retention and loyalty campaigns.

4. Which campaigns perform best?

The Most Recent Campaign recorded the highest response count (334), followed by Campaign 4 (167), Campaign 3 (163), Campaign 5 (163), Campaign 1 (144), and Campaign 2 (30). The business should study what made the recent campaign successful and apply similar targeting strategies.

5. Which customers may be at risk?

Customers with high Recency values and previous meaningful spending should be treated as potential at-risk customers. They can be targeted with personalized discounts, reminders, and retention campaigns before they become inactive.

Key Insights

- Marketing performance should be evaluated at customer, product, channel and campaign levels together.
- High-value customers represent an important opportunity for retention and personalized offers.
- Campaign response analysis can reduce broad, inefficient targeting.
- Purchase-channel behavior can support more relevant digital, catalog and in-store strategies.
- Customer segmentation provides a practical bridge between analysis and marketing action.

Recommendations

- Customer Strategy: Build targeted retention programs for high-value and potentially at-risk customers.
- Product Strategy: Promote high-performing categories while using bundles to increase exposure of lower-performing categories.
- Campaign Strategy: Use historical campaign response to prioritize customers with a higher likelihood of engagement.
- Channel Strategy: Personalize offers according to customers' preferred purchase channels.
- Segmentation: Use RFM or clustering in the next analytical stage to create actionable customer groups.

Power BI Dashboard

The Power BI report is designed as an interactive management dashboard.



Product & Campaign Analysis

Store Purchases

13K

Web Purchases

9K

Catalog Purchases

6K

Overall Campaign Response %

7.45%

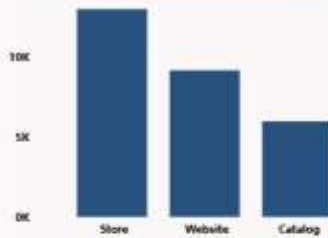
Year

2012

2013

2014

Total Purchase By Channel



Product Contribution %



Accepted Responses by Campaign



Recommendations:

- **Prioritize Store Channel:** Store purchases lead overall, so strengthen in-store promotions and loyalty offers.
- **Focus on High-Contribution Products:** Wines and Meat Products contribute the majority of spending; use bundles and cross-selling to increase basket value.
- **Optimize Campaign Strategy:** Analyze the successful most-recent campaign and replicate its effective targeting or offers.
- **Strengthen Digital Channels:** Website and Catalog purchases trail stores, creating an opportunity for personalized online promotions.

SQL & Data Analysis Work Summary

SQL Analysis

- Customer-level aggregation and profiling.
- Product spending and purchase-channel analysis.
- Campaign acceptance and response calculations.
- Top-customer ranking and comparative analysis.
- Use of CASE statements for customer grouping.
- Use of CTEs and window functions for advanced business analysis.

Python Analysis

- Data cleaning and validation.
- Missing-value and duplicate handling.
- Data type correction and feature engineering.
- Descriptive statistics and group-based analysis.
- Distribution and outlier analysis.
- Correlation analysis and business interpretation.

Power BI Analysis

- Power Query transformations.
- Data modeling and relationships.
- DAX measures and calculated columns where required.
- KPI cards, comparison charts, trend visuals and interactive filters.
- Business-focused dashboard storytelling.

Project Value Delivered

- Converted raw customer marketing data into an analysis-ready dataset.
- Created a reproducible workflow from cleaning through SQL and visualization.
- Identified customer, product, channel and campaign patterns that can support marketing decisions.
- Created a foundation for customer segmentation and targeted marketing optimization.

Project Summary & Next Steps

Key Findings

- Customer spending and purchasing behaviour varies across age, income, education, and household groups.
- Wines contribute the highest share of product spending (50.17%), followed by Meat Products (27.56%).
- Store is the leading purchase channel (13K purchases), followed by Website (9K) and Catalog (6K).
- The most recent campaign recorded the highest response count (334), indicating an opportunity to study its successful targeting strategy.
- Combining recency, total spending, and purchase activity provides a useful foundation for identifying high-value and potentially inactive customers.

Future Scope

RFM analysis, K-Means clustering, campaign-segment comparison, and predictive customer analytics can be added in future work.

Conclusion

The project demonstrates a complete data analytics workflow, starting from raw customer marketing data and progressing through data cleaning, exploratory data analysis, SQL-based business analysis, dimensional modeling, DAX calculations, and interactive Power BI dashboard development.

The final dashboard provides a clear view of customer behaviour, product contribution, purchase channels, and campaign responses. The analysis identified Wines as the largest contributor to product spending, Store as the leading purchase channel, and the most recent campaign as the strongest in terms of response count. These insights can support customer segmentation, product prioritization, campaign optimization, and channel-specific marketing strategies.

Future analysis using RFM and customer clustering can further improve the identification of high-value, loyal, and at-risk customers.